PHY-101

29/08/20

1. A particle in motion has its velocity 'v' at a given time t. If the distance (s) travelled by the particle is expressed as, $s = \frac{1}{2}vt$. show that, the particle is moving with a constant accelaration.

⇒ Given, $s = \frac{1}{2}vt$

$\Rightarrow s = \frac{1}{2}\left(\frac{ds}{dt}\right)t \quad \left[\because v = \frac{ds}{dt}\right]$

$\Rightarrow \int \frac{ds}{s} = 2\int \frac{dt}{t}$ (integrating)

$\Rightarrow \ln s = 2\ln t + \underbrace{\ln c}_{\text{const.}}$

$\Rightarrow \ln\left(\frac{s}{c}\right) = \ln t^2$

$\Rightarrow \boxed{s = ct^2}$

∴ velocity, $v = \frac{ds}{dt} = 2ct$.

∴ accelalation, $\boxed{a = \frac{dv}{dt} = 2c = \text{const}}$

So, acc. is const. here. → proved.

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2. The air bubble formed by explosion inside water perform oscillations with time period T which depends on pressure (p), density (ρ) and on energy due to explosion (E). Establish relation betn T, p, E and ρ.

⇒ Let, $\boxed{T = k\, p^{\alpha} E^{\beta} \rho^{\gamma}}$ (1), with $k, \alpha, \beta, \gamma \rightarrow$ Constants.

we can find α, β, γ by dimensional analysis.

Now $p = \frac{\text{force}}{\text{Area}} = \frac{ma}{\text{area}} = \frac{[MLT^{-2}]}{[L^2]}$

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$\Rightarrow p = [ML^{-1}T^{-2}]$

<u>Energy</u> $E \propto \text{mass}(\text{velocity})^2$

$= [m(LT^{-1})^2]$

$= [ML^2T^{-2}]$

$KE = \frac{1}{2}mv^2$

$E = mc^2$

<u>Density</u>, $\rho = \frac{\text{mass}}{\text{volume}} = \frac{[M]}{[L^3]} = [ML^{-3}]$

From eqn (1) —

$[T]^1 = k[ML^{-1}T^{-2}]^\alpha [ML^2T^{-2}]^\beta [ML^{-3}]^\gamma$

$= k[M^{\alpha+\beta+\gamma}][L]^{-\alpha+2\beta-3\gamma}[T]^{-2\alpha-2\beta}$

~~Comparing~~

$\Rightarrow [M]^0[L]^0[T]^1 = k[M]^{\alpha+\beta+\gamma}[L]^{-\alpha+2\beta-3\gamma}[T]^{-2(\alpha+\beta)}$

<u>Comparing the powers</u>,

$\alpha+\beta+\gamma = 0$

$\Rightarrow \alpha+\beta = -\gamma$ — (2)

$-\alpha+2\beta-3\gamma = 0$

$\Rightarrow -\alpha+2\beta-3\gamma = 0$ — (3)

$-2(\alpha+\beta) = 1$

$\Rightarrow (\alpha+\beta) = -1/2$ — (4)

<u>substituting (2) in (4)</u> $\longrightarrow$

$-\gamma = -1/2$

$\Rightarrow \boxed{\gamma = 1/2}$

(2) — $\alpha+\beta = -1/2$ — (5)

(3) — $-\alpha+2\beta = +3\gamma$

$\Rightarrow -\alpha+2\beta = 3/2$ — (6)

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Solving (5) & (6)

(5) + (6) —

$3\beta = 3/2 - 1/2 = 1$

$\Rightarrow \boxed{\beta = 1/3}$

(5) $\longrightarrow \alpha + 1/3 = -1/2$

$\Rightarrow \alpha = -1/2 - 1/3 = -\left(\frac{3+2}{6}\right) = -5/6$

So:

we get the relation —

$T = k\, p^{-5/6} \cdot E^{1/3}\, \rho^{1/2}$

$$\boxed{T = k \frac{\rho^{1/2}\, E^{1/3}}{p^{5/6}}}$$ (Ans)

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3. A special effects studio wants to film a scene in which the leading Tower of Pisa (take height, h = 100 m) topples to the ground. In order to achieve this, the studio might make a scale model of the tower, which is (say) 1 m tall, and then film the model falling over. The only problem is that the resulting footage would look completely unrealistic because the model tower would fall over too quickly. The studio could easily fix this problem by slowing the film down. The question is by what factor should the film be slowed down in order to make it look realistic? Take 'm' as the mass of the tower and 'g' as the acc. due to gravity.

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⇒ Let, us assume that the time taken for the fall depends upon, mass, height of the tower and acc. due to gravity.

i.e, $\boxed{t \propto m^{\alpha} h^{\beta} g^{\gamma}}$ —— (1)

By dimensional analysis,

$[t] = k[m^{\alpha} h^{\beta} g^{\gamma}]$, $k \rightarrow$ constant.

$\Rightarrow [T]^1 = k[M]^{\alpha}[L]^{\beta}[LT^{-2}]^{\gamma}$

$\Rightarrow [T]^1 = k[M]^{\alpha}[L]^{\beta+\gamma}[T]^{-2\gamma}$

Comparing, the powers on both sides,

$\alpha = 0$ | $\beta + \gamma = 0$ | $-2\gamma = 1$

$\Rightarrow \boxed{\beta = -\gamma}$ $\Rightarrow \boxed{\gamma = -1/2}$

$\Rightarrow \boxed{\beta = 1/2}$

So, $\boxed{t = k h^{1/2} g^{-1/2}}$ —— (2)

Let, t_1 & h_1 represents the time of fall & height of actual tower. & t_2 & h_2 be that of the model.

So, using (2),

$$\frac{t_1}{t_2} = \frac{h_1^{1/2}}{h_2^{1/2}} \frac{\cancel{g^{-1/2}}}{\cancel{g^{-1/2}}}$$

$h_1 = 100m$, $h_2 = 1m$ } given

$\Rightarrow \frac{t_1}{t_2} = \left(\frac{100}{1}\right)^{1/2} = 10$

$\Rightarrow \boxed{t_1 = 10 t_2}$

So, actual tower takes 10 times more time to fall compared to the model.

So, to make the movie more realistic the film should be slowed down by a factor of 10.

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4. Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon the velocity v of the water, it's density 'ρ' and acc. due to gravity. Show, that m varies as the sixth power of the velocity of water in the river.

$\Rightarrow$ Let, $\boxed{m = k\, v^{\alpha} \rho^{\beta} g^{\gamma}}$, $k \rightarrow$ const. —— (1)

By dimensional analysis,

$$[M] = k \cdot [LT^{-1}]^{\alpha} [ML^{-3}]^{\beta} [LT^{-2}]^{\gamma}$$

$$\cancel{\Rightarrow [M] = k\, [L]^{\alpha - 3\beta + \gamma}}$$

$$\Rightarrow [M] = k\, [M]^{\beta} [L]^{\alpha - 3\beta + \gamma} [T]^{-\alpha - 2\gamma}$$

$$\boxed{\begin{aligned} v &= \frac{s}{t} \\ \rho &= \frac{m}{V} \\ g &= \frac{v}{t} = \frac{s/t}{t} \end{aligned}}$$

Comparing powers on both side –

$\boxed{\beta = 1}$

$\alpha - 3\beta + \gamma = 0$

$\Rightarrow \alpha + \gamma = 3\beta$

$\Rightarrow \alpha + \gamma = 3$ —— (2)

$-(\alpha + 2\gamma) = 0$

$\Rightarrow \alpha + 2\gamma = 0$ —— (3)

(3) – (2) $\rightarrow$ $\boxed{\gamma = -3}$

from (2) –

$\cancel{\alpha = 3 + \gamma}$ $\alpha = 3 - \gamma$

$= 3 + 3$

$\boxed{\alpha = 6}$

from (1) –

$$\boxed{m = k\, v^{6} \rho\, g^{-3}}$$

so, $\boxed{m \propto v^{6}}$ (proved)

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Q: Convert the point $(-1, 1, -\sqrt{2})$ from cartesian to spherical co-ordinates.

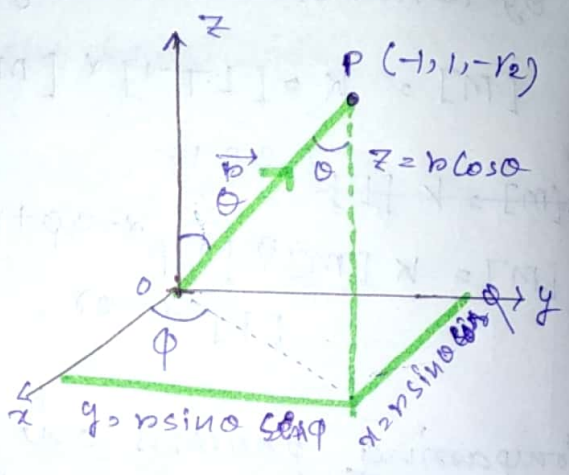
⇒ ~~Relation betn cartesian & spherical co-ordinates~~ –

We have to convert →

$$(x, y, z) \longrightarrow (r, \theta, \phi)$$

We know the relations betn cartesian co-ordinate & spherical co-ordinate as –

$$x = r \sin\theta \cos\phi$$
$$y = r \sin\theta \sin\phi$$
$$z = r \cos\theta$$



where

$$\left\{\begin{aligned} r^2 &= (x^2 + y^2 + z^2) \\ \cos\theta &= z/r \\ \sin\phi &= \frac{y}{r} \cdot \frac{1}{\sin\theta} \\ \phi &= \tan^{-1}(y/x) \end{aligned}\right\}$$

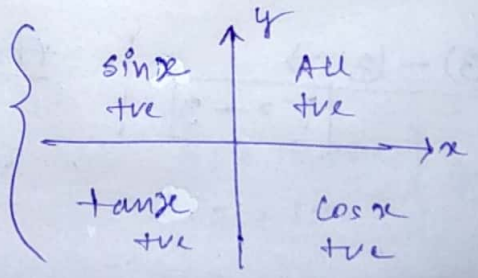
$$\left\{\begin{aligned} x &= -1 \\ y &= 1 \\ z &= -\sqrt{2} \end{aligned}\right\}$$

Now $r^2 = (1 + 1 + 2) = 4$

$\Rightarrow \boxed{r = 2}$

$\therefore \cos\theta = \frac{-\sqrt{2}}{2} = -1/\sqrt{2}$

So, $\theta = (\pi/2 + \pi/4)$ & $(\pi + \pi/4)$



$$\boxed{\theta = 3\pi/4,\ 5\pi/4}$$

~~Now, sinφ = y/r~~.

Now, as $\left.\begin{aligned} x &= -1 \\ y &= 1 \end{aligned}\right\}$ so, the point is in the 2nd quadrant.

So, θ must be $= 3\pi/4$ $\Rightarrow \boxed{\theta = 3\pi/4}$

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$\therefore \sin\phi = \frac{y}{r} \frac{1}{\sin\theta}$

$= \frac{1}{2} \frac{1}{\sin(3\pi/4)}$

$= \frac{1}{2} \cdot \frac{1}{1/\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow \boxed{\phi = \pi/4, 3\pi/4}$

As the particle is in the 2nd quandant, so, ϕ must be $= 3\pi/4$

So, $\boxed{(r, \theta, \phi) = (2, 3\pi/4, 3\pi/4)}$

$* \cdot \frac{y}{x} = \frac{r\sin\theta \sin\phi}{r\sin\theta \cos\phi} = \tan\phi$

$\Rightarrow \boxed{\phi = \tan^{-1}(y/x)}$

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